

Predicted Failure Load (kN)

**XGBoost
hold-out test
MAPE=3.18%
 $R^2=0.994$**

30
25
20
15
10
5

5

10

15

20

25

30

Actual Failure Load (kN)

- **Hold-out test (real data)**
- **Perfect prediction**
- ±10% error band**

